

Department of Data Science and Artificial Intelligence
National Institute of Technology Kurukshetra
B.Tech (AIML), AI/PC-205/DBMS, Mid Term (II) Exam, Odd Sem 2025-26

Time Allowed: 50 Mins
Max. Marks: 15.00

Date of Exam: 12.11.2025
Roll No: 224103055

Note: All the questions and their parts are compulsory; there is no choice. Answer each part of a question in the same place.

Q1.	Anwar the following briefly: (a). Why SQL does not allow to compare an attribute value to NULL using '$=$' and '$<$' operators? How this problem is resolved in SQL? (b). In DB design, when it is required to derived the Super Key (s)? List the utility of Super key (s) in DB normalization. (c). Consider a hospital relation <i>PATIENT_VISIT</i> (<i>VisitID</i>, <i>PatientID</i>, <i>DoctorID</i>, <i>Department</i>, <i>Treatment</i>, <i>Cost</i>). The known FDs are: <i>VisitID</i> $\rightarrow$ <i>PatientID</i>, <i>DoctorID</i>, <i>Treatment</i>, <i>Cost</i>, <i>DoctorID</i> $\rightarrow$ <i>Department</i>, <i>PatientID</i>, <i>Treatment</i> $\rightarrow$ <i>Cost</i> (Same treatment costs same for patient). Using inference rules, prove that: Can we infer <i>DoctorID</i> $\rightarrow$ <i>Cost</i>? Why or why not?	1*3
Q2.	Consider the <i>Company</i> relational DB, where the primary key(s) are underlined. <i>employee</i> (<u><i>employee-name</i></u>, <i>gender</i>, <i>salary</i>, <i>city</i>, <i>country</i>) <i>company</i> (<u><i>company-name</i></u>, <i>city</i>, <i>country</i>) <i>works</i> (<u><i>employee-name</i></u>, <u><i>company-name</i></u>) <i>manages</i> (<u><i>employee-name</i></u>, <u><i>manager-name</i></u>) (a) Give an equivalent Relational Algebra expression for each of the following queries: 2.00 (i) Find employees who manage everyone in their city (ii) Find employees who earn more than their manager 4.00 (b) Give an equivalent SQL expression for each of the following queries: (i) Delete employees who don't work for any company. (ii) Find the names of all female managers who work for the company named 'IBM' but do not live in any city in the 'USA'. (iii) List managers who earn less than someone they manage. (iv) For Company expansion analysis: find cities with companies but no employees living there.	
Q3.	(a) Consider a relation <i>R</i> (<i>A</i>, <i>B</i>, <i>C</i>, <i>D</i>, <i>E</i>, <i>F</i>), with FDs: $\{A \rightarrow BC, D \rightarrow EF, C \rightarrow A, F \rightarrow D\}$ and Decomposition: $D1 = \{R1(A, B, C), R2(C, D, E, F), R3(C, D)\}$. Check if the decomposition is dependency preserving or not. Also check whether it is Lossless or lossy decomposition. Discuss Briefly. 2.00 (b) Consider relation <i>R</i>(<i>A</i>, <i>B</i>, <i>C</i>, <i>D</i>, <i>E</i>, <i>F</i>) and FDs: $\{AB \rightarrow C, C \rightarrow D, D \rightarrow E, E \rightarrow F, F \rightarrow A\}$. Answer the below: 4.00 (i). Identify all candidate keys. (ii). Determine the highest normal form (1NF, 2NF, 3NF, BCNF) that the relation is in. Justify your answer by showing which, if any, dependencies violate each normal form.	